MTH 201 Chapter 2: Integration Techniques

2.5 Worksheet

1. Entry #95 in the table of integrals states $\int \frac{u^2}{\sqrt{u^2+a^2}}\,du = \frac{u}{2}\sqrt{u^2+a^2} - \frac{a^2}{2}\ln\left(u+\sqrt{u^2+a^2}\right)+C$. Use this fact to determine the integral $\int \frac{2x^2}{\sqrt{2x^2+4}}\,dx$.

2. Entry #117 in the table of integrals states $\int \frac{du}{u\sqrt{2au-u^2}} = -\frac{\sqrt{2au-u^2}}{au}+C$. Use this fact to determine the integral $\int \frac{dx}{x\sqrt{6x-x^2}}$.

3. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{2x}{3x+4}\,dx$

4. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\ln(x)}{\sqrt{x^3}}\,dx$.

5. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{x}{(4-5x)^2}\,dx$.

6. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\sqrt{3x^2-5}}{3x^2}\,dx$.

7. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int \frac{\cos(4x)}{9+\sin^2(4x)}dx$.

8. Determine an entry in the table of integrals that can be used to evaluate the integral: $\int x^3\sqrt{5-9x^2}\,dx$.